

eye and slices of the left-side perspective reach his left eye.

The benefit of this technology is that it does not require the viewer to sit directly in front of the screen. New TV screens use several sets of lenses to create multiple left and right image pairs for viewers sitting at different angles from the TV. One set of images is for viewers seated directly in front of the screen. Another is for viewers to the side of the screen. Additional pairs take care of all the viewers in between. If you move sideways, you simply transition from one pair of right-left images to the next.

This technology does not require you to wear glasses but the 3D experience is nowhere close to that of active 3D technology.

In smartphones: autostereoscopy

Autostereoscopy is the process of displaying 3D images or stereo images without using specialised headgear. It is also known as glasses-free 3D and is the major technology used in 3D smartphones such as HTC EVO 3D and LG Optimus 3D, and portable gaming devices such as the Nintendo 3DS.

This technology uses the principle of parallax barrier to deliver different images to each eye. The latest and most convenient design does not have the physical parallax barrier in front of the pixels, but behind the pixels and in front of the backlight. Thus it doesn't send different images but different light to the two eyes. This allows the two channels of light to pass through the pixels, allowing glare over the opposite pixels, giving the best image quality.

What's next?

A different TV programme for everyone. 3D technology is being tested for use in such a way that we can build devices that show different images or videos to different people looking at the same screen. Currently, 3D devices send different images to the two eyes. With a fair amount of tweaking, it can also be possible that these devices send



Nintendo 3DS portable gaming device

different images to different pairs of eyes. For instance, in passive 3D TV, the angle at which the viewer sits away from the TV set decides the pair of right-left images he sees. If different pairs of right-left images carry different programmes, we have the ultimate solution to the age-old family problem of fighting over the remote control.

3D conferencing. 3D videoconferencing allows seamless transmission of 3D media across the network, revolutionising traditional video conferencing through introduction of more reality using customised 3D technology.

The key advantage that 3D video conferencing brings against traditional voice over Internet protocol (VoIP) solution is superior visual aspect and vivid perception of facial and other body language, resulting in better involvement of users and enhanced overall communication. Best of all, this technology allows for eye contact.

3D processing hardware. High-end enthusiast laptops and consumer laptops that incorporate 3D capability have been available for the last one year. While mainstream laptops that incorporate 3D use the Nvidia GT 555M GPU, enthusiast devices such as the Alienware M17xR3 use higher-end cards like the Nvidia GTX 580M. In fact, most current GTX GPUs support 3D gaming.

A 120Hz monitor and a 3D Vision Kit combined with a compatible Nvidia graphics card transform the mundane 2D images on the PC into a realistic 3D experience. 3D Vision requires proper configuration before the user can really enjoy the effect. The technology degrades the performance in most games by 50 per cent. Also, the shutter glasses can be annoying to wear for a long time and block a lot of the display brightness. Nevertheless, Nvidia is still in the lead compared to AMD.

AMD uses a technology called 'HD3D' to enable stereoscopic 3D support in games, movies and/or photos. Additional hardware (such as 3D-enabled panels, 3D-enabled glasses/emitter and Blu-ray 3D drive) and/or software (such as Blu-ray 3D disks, 3D middleware and games) are required for enabling stereoscopic 3D.

The Radeon HD cards such as the AMD Radeon 5450 and the 6970 HD cards also support 3D gaming. Companies that use these cards are MSI, Lenovo and Hewlett Packard (HP). The HP Envy 17 3D is one of the most popular PCs to come with the Radeon 3D solution. Although AMD has a higher-end card called the Radeon DH 6990, AMD Crossfire Technology's incompatibility with AMD's Quad Buffer causes it to not support stereoscopic 3D gaming with frame-sequential displays using active shutter glasses.

Ultimately, most high-end GPUs can play 3D games, but as the overload is almost double the frame-per-second of the game, play performance will probably halve. Moreover, due to software incompatibility, certain performance boosting setups such as AMD Crossfire cannot be used. This puts a cap on the maximum gaming performance that can be derived from those cards.